

Emmanuel Basua — Resume

Electrical & Computer Engineering | Cape Town, South Africa

Profile

Electrical & Computer Engineering student at the University of Cape Town focused on embedded systems, signal processing, control and practical hardware prototyping.

Education

University of Cape Town — Electrical & Computer Engineering.

Experience

- Embedded Systems Tutor — University of Cape Town (2025): digital logic and microcontroller programming in C.
- Engineering Design Tutor — University of Cape Town (2024): PCB design in KiCad and introductory engineering design.
- UCT Formula One Team — Electrical (2024): research into the Brake System Plausibility Device and Brake Over Travel Switch.
- SHAWCO Volunteer (2023–): digital literacy teaching in Cape Town.

Technical Skills

- Embedded C / C++ / ESP32 / Raspberry Pi / PX4
- Python / MATLAB / Simulink / DSP / Flask
- KiCad / CAD / Linux / Git / electronics bench work
- Sensor fusion / control systems / telemetry / rapid prototyping

Selected Projects

- Gyro-Enhanced Micromouse
- AUV Acoustic Communication
- Rocket Telemetry